

Module 49 - Deep Copy vs Shallow Copy, Call by Value & Reference

1. What is Call by Value?

Call by Value means a copy of the value is passed or assigned to another variable, so changing one variable does not affect the other variable.

Example Code:

```
// Stack - to store primitive values
// Heap - to store non-primitive values (array, object, function)

/* Call By Value

Stack (Before Modification):

| Address | Variable | Value |
|-----|-----|-----|
| 0x100   | x       | 10    |
| 0x104   | y       | x = 10 |

Stack (After Modification):

| Address | Variable | Value |
|-----|-----|-----|
| 0x100   | x       | 10    |
| 0x104   | y       | 20    |

*/
let x = 10;
// The value of x is copied into y.
let y = x;
// Changing y does not affect x because y has its own copied value.
y = 20;
console.log(x, y);
```

2. What is Call by Reference?

Call by Reference means multiple variables refer to the same object or array in memory, so changing the object or array through one variable also affects the other variable.

Example Code:

```
/* Call By Reference
```

Stack Memory:

Address	Variable	Reference
0x100	obj1	0x001
0x104	arr1	0x002
0x108	obj2	0x001
0x112	arr2	0x002

Heap Memory:

Address	Object
0x001	{name : 'Kesavan'}
0x002	[1, 2, 3]

```
*/
```

```
let obj1 = {  
  name: "Kesavan"  
};  
let arr1 = [1, 2, 3];  
// obj2 receives the same reference as obj1.  
let obj2 = obj1;  
// arr2 receives the same reference as arr1.  
let arr2 = arr1;
```

```
// Changes the object through the obj1 reference.  
obj1.name = "Murugesan";  
obj1.role = "Developer";  
  
// Changes the array through the arr1 reference.  
arr1[0] = "One";  
console.log(obj1, obj2);  
console.log(arr1, arr2);
```

3. What is Shallow Copy?

Shallow Copy creates a new object or array, but nested objects or arrays inside it still share the same reference with the original.

4. What is Deep Copy?

Deep Copy creates a completely independent copy, including all nested objects and arrays, so changes in the copy do not affect the original.

Example Code:

```
// Primitive Copy  
let a = 10;  
let b = a;  
a = 20;  
console.log(a, b);  
  
// Object - Shallow Copy  
let obj1 = {  
  name1 : "Murugesan",  
  role : "Developer"  
};  
let obj2 = {...obj1};  
obj2.name1 = "Kesavan";  
console.log(obj1, obj2);
```

```
// Nested Object
```

```
let person = {  
  name1 : "Johny",  
  role : "Youtuber",  
  hobbies : {  
    cricket : "T20 Player",  
    football : "60 mins"  
  }  
};
```

```
// Shallow Copy with Nested Object
```

```
let person2 = {...person, hobbies : {...person.hobbies}};
```

```
// Deep Copy with Nested Object
```

```
let newObj = JSON.parse(JSON.stringify(person));  
person.role = "Video Editor";  
person.hobbies.cricket = "ODI Player";  
console.log(person, person2);  
console.log(newObj);
```

```
// Nested Array
```

```
let Arr = [1, 2, 3, [4, 5, 6]];
```

```
// Shallow Copy with Nested Array
```

```
let Arr1 = [...Arr];
```

```
// Deep Copy with Nested Array
```

```
let newArr = JSON.parse(JSON.stringify(Arr));  
Arr[0] = "one";  
Arr[3][0] = "Four";  
newArr[3][1] = "Five";  
console.log(Arr, Arr1);  
console.log(newArr);
```